Sofija Dimitrijevic

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Education

University of California, Santa Barbara Sep. 2026 – Jun. 2028
M.S. in Computer Science, Machine Learning Focus

California Polytechnic State University, San Luis Obispo Sep. 2022 – Jun. 2026
B.S. in Computer Science, AI/ML Concentration; Minor in Linguistics GPA: 3.8

- **Magna Cum Laude**; Dean's List all 12 quarters; President's Honor List 2022–2023 and 2023–2024; Society of Women Engineers Scholarship, 2025 & 2026.
- **Relevant coursework:** Speech & Language Processing, Deep Learning, Machine Learning, Artificial Intelligence, Operating Systems, Software Engineering, Database Systems, Distributed & Parallel Computing, Theory of Computation, Computer Networks, Computer Security, Applied Regression Analysis, Data Structures & Algorithms, Object-Oriented Programming in Java, Systems Programming.

Research Experience

Research Assistant Aug. 2025 – Present
Cal Poly Institute For Advanced Technology and Public Policy San Luis Obispo, CA

- Developed machine learning methods for speaker identification in legislative testimony through the CalMatters Digital Democracy project, using a corpus of **1.54 million utterances** across five state legislative sessions.
- **Original paper:** Coauthored *Detection of Self-Introductions in Legislative Testimony* with Foaad Khosmood [2]. Built an automatically labeled training dataset using name-matching heuristics and engineered lexical, positional, structural, and introductory-phrase features; compared decision tree, random forest, and XGBoost classifiers, achieving **0.9704 F1** with XGBoost.
- **Expanded manuscript:** First author of an expanded version of *Detection of Self-Introductions in Legislative Testimony*, with Pallavi Das, Kasey Liu, and Foaad Khosmood [1]. Extended the feature set with discourse context and fine-tuned BERT probability features; the hybrid XGBoost model improved F1 from **0.9747 to 0.9782** and reduced test errors from **241 to 207**.
- Analyzed classification errors and heuristic label noise, identifying name inconsistencies and self-introductions without explicit introductory phrases as sources of evaluation and modeling challenges.

Research Assistant Nov. 2023 – Jun. 2025
Tab Lab Physics, Cal Poly Physics Department San Luis Obispo, CA

- Integrated novel flocking dynamics into the Active Finite Voronoi (AFV) model to study collective cell motion in cancer progression and wound healing.
- Vectorized critical kernels, **improving simulation runtime by 50%**; migrated model's legacy MATLAB codebase to Python for better maintainability and HPC scalability.
- Selected as summer research intern in 2024 for Boston research residency, collaborated with Northeastern Center for Theoretical Biophysics to improve simulation efficiency and real-world applicability.

Industry Experience

Software Engineering Intern Jun. 2026 – Sep. 2026
Microsoft Redmond, WA

- Shipped a production **AI agent** to debug live-site incidents for Microsoft's Commercial Pricing service, diagnosing hundreds of incidents in a **median 4.8 minutes** from alert to written root cause. Authored **5 diagnostic skills** and grounded the agent in **6 live data sources**.
- Scaled the agent into a **multi-subagent system** with autonomous incident resolution, closing dozens of incidents with **zero reactivations** and increasing 24-hour resolution of predictable failures from **20.5% to 35.5%**.
- Built an automated weekly evaluation report to distinguish agent failures from gaps in skill coverage. Identified that approximately **70% of incidents were repeats**, with authored skills handling **30% of those repeats**, informing the team's automation priorities.

Software Engineering Intern Jun. 2025 – Sep. 2025
HP Inc. Houston, TX

- Designed and deployed an **AI-powered telemetry service** with a Random Forest model, **boosting prediction accuracy by 50%** and reducing hardware validation costs.
- Built a C#/.NET desktop client with IPC messaging for **real-time inference at scale**, enabling deployment across thousands of devices. Created **modern UI** for application using WPF and Caliburn.
- Automated performance benchmarking with SQLite-backed data storage, **reducing validation time by a factor of three** and improving reproducibility across teams.

Research Papers

[1] **Dimitrijevic, S.**, Das, P., Liu, K., & Khosmood, F. (2026). *Detection of self-introductions in legislative testimony.* Expanded study presented at AAIRC, Ai4 2026. arXiv:2608.07891 [cs.CL]. doi:10.48550/arXiv.2608.07891.

[2] **Dimitrijevic, S.**, & Khosmood, F. (2026). *Detection of self-introductions in legislative testimony.* Original manuscript presented at the Computer Science Conference for CSU Undergraduates (CSCSU 2026). Unarchived manuscript.

Research Posters & Presentations

[Presentation] *Detection of Self-Introductions in Legislative Testimony* (expanded study with BERT) [1]. Sofija Dimitrijevic, Pallavi Das, Kasey Liu, and Foaad Khosmood. Applied AI Research Conference (AAIRC), Ai4 2026, The Venetian, Las Vegas, NV. *August 3, 2026.*

[Presentation] *Detection of Self-Introductions in Legislative Testimony* (original study) [2]. Sofija Dimitrijevic and Foaad Khosmood. Computer Science Conference for CSU Undergraduates (CSCSU 2026), virtual. *April 25, 2026.*

[Poster] *Introducing Orientational Alignment to Simulate Flocking Behavior in Active Finite Voronoi Model.* Cal Poly Bailey College of Science and Mathematics 2025 Research Conference *May 2025.*

[Poster] *Visual Perception in an Agent-Based Model of Predator-Prey Relationships.* Cal Poly Bailey College of Science and Mathematics 2024 Research Conference *May 2024.*

Selected Projects

Spotify Analyzer & Recommender — *Python, scikit-learn, Pandas, Matplotlib* Mar. 2025

- Collected and processed **7M+ streaming records** from 54 Spotify users (2012–2025) to explore music-taste patterns and build personalized recommenders.
- Engineered multiple sparse matrices (user–song, user–artist) and implemented **user-based weighted collaborative filtering** to generate song and artist recommendations, tuning neighbor count for best results.
- Applied **K-Means and DBSCAN** clustering to detect "niche" listening habits and identify universal artists, visualizing artist clusters and user similarity graphs (cosine and Pearson distance performed best).
- Built a **K-Nearest-Neighbors classifier** to predict a listener's **home region** from music preferences, highlighting geographic taste trends.

Gather: Inventory Management Web App — *React, TypeScript, REST API* Apr. 2024 – Jun. 2024

- Built full-stack inventory management system with REST APIs and CRUD features in a 5-person Agile team.
- Refactored backend into modular services, reducing code duplication by **50%** and supporting CI testing pipelines.

Moment: Burnout Prediction (Camp Poly Hacks Hackathon) — *Python, scikit-learn, Figma* Jan. 2024

- Developed ML models (linear regression, decision trees) for predicting hospital staff burnout with an F1 score of 0.80; designed app UI and logo, collaborated on web app prototype.
- Contributed in a team of 5 to deliver prototype in 24 hours; won **Best Design** award out of 13 teams. See the hackathon showcase.

Additional Experience, Mentorship & Outreach

Women Involved in Software & Hardware (WISH) 2022 – 2026
Vice President; Technical Director (2024–2025); Tech Officer (2023–2024) Cal Poly

- Led operations for a **300+ member** community; managed a 20-person officer board.
- Coordinated annual Grace Hopper trip logistics and pre-conference prep sessions (networking, resumes, outreach).
- Organized 20+ annual build days and workshops (soldering, breadboarding, web dev, SQL, CTFs, etc.); on-boarded members to software/hardware fundamentals; mentored officers and workshop attendees.

CSC 307/308, CSSE Department 2025 – 2026
Grader Cal Poly

- Graded assignments and projects for upper-division Software Engineering courses.

ENGL 391, English Department Fall 2025
Teaching Assistant Cal Poly

- Held weekly office hours to support students and graded assignments for an Applied Linguistics course.

Awards & Honors

- SWE Scholarship Recipient (2025 & 2026).
- Magna Cum Laude, B.S. in Computer Science (2026); Dean's List all 12 quarters; President's Honor List 2022–2023 and 2023–2024.

Professional Conferences

Grace Hopper Conference 2025 — Attendee (*Nov. 2025*).

Society of Women Engineers WE 24 — Attendee (*Oct. 2024*).

Harvard WECode — Attendee (*Feb. 2024*).

Certifications & Training

- Anthropic Introduction to Model Context Protocol (Aug. 2025).
- Anthropic Advanced Model Context Protocol (Dec. 2025).
- Codepath Advanced Interview Prep Course (Jun. - Sep. 2024).

Technical Skills

Languages: Python, Java, C/C++, C#, SQL, KQL, R, MATLAB, JavaScript, TypeScript, HTML/CSS, Racket
AI/Agents: LLM agents, MCP, agent evaluation, agent safety, prompt engineering, retrieval/RAG.
Cloud/Infrastructure: Azure (DevOps, Data Factory, EV2), Bicep/ARM, CI/CD, Docker, Linux, Git.
Frameworks/Tools: scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Optuna, SciPy, spaCy, NLTK, PyTorch, TensorFlow, Hugging Face, CUDA, .NET, WPF, AsyncIO, MCP, React, Node.js, Express.js, MongoDB, Git, CI/CD, Docker, Jupyter, Google Colab, Spark, SQLite, Jest, Pytest, Linux, Figma.

Additional

Interests: Mentoring in women-in-tech communities, computational linguistics, tennis.